

ANKESH SINGH

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LinkedIn

GitHub

Portfolio

EDUCATION

Indian Institute Of Information Technology Agartala

Bachelor of Technology in Computer Science and Engineering

2023 – 2027

CGPA: 8.52

Amrit Public School Mau, UP

Senior Secondary (XII)

2021 – 2022

Percentage: 90.0%

EXPERIENCE

Bureau of Indian Standards (BIS)

May 2026 – July 2026

Software Engineering Intern

- Developed a dual-camera **computer vision system** to automate monitoring of **Lever Device** tests for water bottles as per **IS 17803**, eliminating continuous manual observation during the **1,500-cycle, 70-minute test**.
- Implemented **OCR-based real-time counter monitoring** for machine cycle tracking and automated **Pass/Fail decisions** based on detected anomalies and target counter values.
- Built **hinge and lid rupture detection** using ROI-based image processing techniques to automatically identify test failures.
- Created a **multithreaded PyQt5 application** with live monitoring, Pass/Fail alerts, and automated report generation.
- Technologies:** Python, PyQt5, OpenCV, Tesseract OCR, ReportLab, Multithreading

PROJECTS

ALEX — AI Knowledge & Research Assistant

- Engineered a **document-grounded RAG pipeline** using **LangGraph**, **FastAPI**, and **Google Gemini**, enabling contextual question answering with conditional retrieval and fallback handling when relevant context is unavailable.
- Implemented semantic document retrieval using **FastEmbed** with **BAAI/bge-small-en-v1.5** embeddings and **PostgreSQL with pgvector**, using cosine-distance search for relevant document chunks.
- Built an end-to-end document processing pipeline supporting **3 file formats**—PDF, DOCX, and TXT—with automated chunking, **384-dimensional embeddings**, page-level metadata, and structured citations containing source documents, pages, and relevance scores.
- Tech Stack:** Python, FastAPI, LangGraph, Google Gemini, FastEmbed, PostgreSQL, pgvector, React.js, Vite, Tailwind CSS, Docker, Railway, Vercel.

Food Distribution Platform — MERN Stack Web Application

- Engineered a full-stack **MERN** platform connecting food donors with NGOs and shelters, implementing donation, claim, and pickup workflows with **JWT authentication**, **refresh tokens**, and **RBAC** across **3 roles**—Donor, Recipient, and Admin.
- Designed and integrated **19 REST API endpoints** using Node.js, Express.js, MongoDB, and Mongoose for authentication, donations, claims, profiles, and administrative operations, secured with input validation, rate limiting, Helmet, and CORS.
- Built real-time donation tracking using **Socket.io** with **4 workflow states**—Available, Claimed, Picked-up, and Completed—and integrated **Google Maps**, **Twilio SMS**, and **Nodemailer** for location-based matching and automated notifications.
- Tech Stack:** MongoDB, Express.js, React.js, Node.js, Socket.io, JWT, Mongoose, Material-UI, Google Maps API, Twilio, Nodemailer.

CERTIFICATIONS & ACHIEVEMENTS

- Meta Hacker Cup 2024:** Advanced to **Round 2** of Meta's global programming competition, ranking **2,521st** in Round 1 and **3,636rd** globally in Round 2.
- LeetCode:** Solved **715+ problems** across Easy, Medium, and Hard categories with a peak contest rating of **1792+**. [Profile](#)
- CodeChef:** Achieved a peak contest rating of **1424+ (2-Star)** through regular participation in competitive programming contests. [Profile](#)
- GirlScript Summer of Code (GSSoC):** Contributed to **open-source projects** by implementing features, resolving issues, and collaborating through Git and GitHub.

TECHNICAL SKILLS

Languages: C++, C, JavaScript, Python

AI & Machine Learning: RAG, LangGraph, Google Gemini, FastEmbed, Embeddings, Vector Search

Frontend: React.js, HTML5, CSS3, Tailwind CSS, Vite

Backend & APIs: FastAPI, Node.js, Express.js, RESTful APIs, Socket.io

Databases & Tools: PostgreSQL, pgvector, MongoDB, MySQL, Mongoose, Git, GitHub, Postman

Cloud & DevOps: Docker, Railway, Vercel, Linux Shell, GitHub Actions

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks